Lecture 1

Uniform limits and fixed points

Objectives. *Physics uses limits inside integrals, derivatives and equations. This lecture establishes uniform bounds that justify these operations. Completeness turns successive approximations into a solution. The contraction principle then gives existence, uniqueness and an error bound.*

By the end of this lecture you will be able to:

- Distinguish pointwise convergence from uniform convergence.
- Check the hypotheses for integrating or differentiating a limit.
- Differentiate a power series inside its convergence disk.
- Use the contraction principle with an explicit error bound.

1.1 Three limit exchanges

For functions $f_n : K \to \mathbb{C}$, *pointwise convergence* to f means $f_n(x) \to f(x)$ for every fixed $x \in K$. The required index can depend on x. Pointwise convergence alone does not justify integrating a limit. All integrals in this lecture are Riemann integrals.

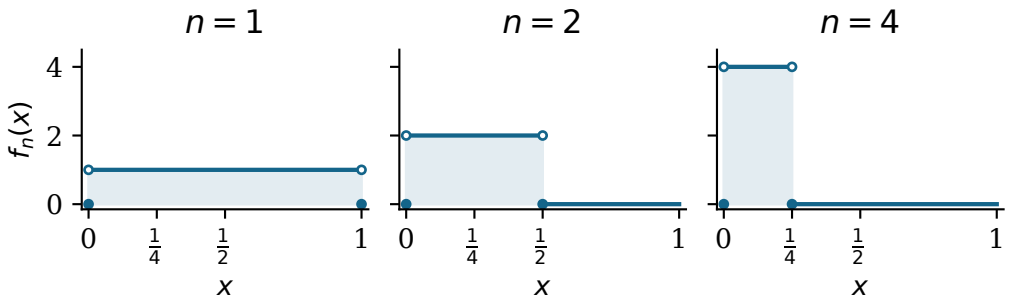
The indicator $\mathbb{1}_A$ equals 1 on a set A and 0 outside it. On $[0,1]$, consider

$$f_n(x) = n\mathbb{1}_{(0,1/n)}(x), \qquad n \geq 1.$$

For every fixed $x \in [0,1]$, the value $f_n(x)$ is eventually zero. Nevertheless,

$$\int_0^1 f_n(x)\,\mathrm{d}x = 1, \qquad \int_0^1 \lim_{n\to\infty} f_n(x)\,\mathrm{d}x = 0.$$

The shrinking rectangles show how the nonzero integral persists.



Each rectangle has height n and width $1/n$. Its area is one, although its nonzero interval shrinks towards zero. The panels show $n = 1, 2, 4$. Open markers indicate excluded endpoints.

Differentiation needs its own control. Consider the series

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin(n^2 x)}{n^2}, \qquad x \in \mathbb{R}.$$

For every integer $N \geq 1$, its tail has a bound independent of x:

$$\left| \sum_{n>N} \frac{\sin(n^2 x)}{n^2} \right| \leq \sum_{n>N} \frac{1}{n^2} \leq \frac{1}{N}.$$

The last inequality follows by comparison with $\int_N^\infty t^{-2}\,\mathrm{d}t$. Yet the derivative of each summand at zero is 1, so the proposed derivative series diverges there. No conclusion about the derivative of f itself is needed for this counterexample. The derivative series cannot be justified by the bound on the original series.

A valid exchange will give a series for a Gaussian integral:

$$\int_0^1 e^{-x^2}\,\mathrm{d}x = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(2n+1)}. \tag{1.1}$$

We justify this identity in § 1.3, using the familiar real exponential series. The integral, the derivative and the infinite sum therefore require separate hypotheses.

To choose a bound, first name the operation being exchanged with the limit. For integration on a bounded interval, a uniform bound on the function error controls the integral error. For differentiation, we instead control the derivative error and fix the function value at one point. For a fixed-point iteration, we first check that the map preserves the chosen complete space. Only then does the contraction estimate prove that the iterates converge.

1.2 Completeness in the supremum norm

Recall the normed-space vocabulary of Linear Algebra for Physics, Lecture 13. A *norm* on a real or complex vector space assigns a nonnegative number $\|u\|$ to each vector and satisfies

$$\|u\| = 0 \iff u = 0, \qquad \|cu\| = |c|\,\|u\|, \qquad \|u+v\| \leq \|u\| + \|v\|.$$

A sequence is *Cauchy* if for every $\varepsilon > 0$ there is an index N such that $\|u_n - u_m\| < \varepsilon$ whenever $n, m \geq N$. A normed space is *complete*, or a *Banach space*, if every Cauchy sequence has a limit in the space.

Let $K \subset \mathbb{R}^d$ be nonempty and compact. Write $C(K)$ for the continuous complex-valued functions on K. Their *supremum norm* is

$$\|f\|_\infty = \sup_{x \in K} |f(x)|.$$

The supremum is finite by compactness. The scalar triangle inequality proves the triangle inequality for this norm. For functions on any nonempty set S, *uniform convergence* means that for every $\varepsilon > 0$ there is an index N such that $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq N$ and $x \in S$. Equivalently, $\sup_{x \in S} |f_n(x) - f(x)| \to 0$. A sequence is *uniformly Cauchy* if, for every $\varepsilon > 0$, one index N makes $|f_n(x) - f_m(x)| < \varepsilon$ for every $m, n \geq N$ and $x \in S$. On $C(K)$, uniform convergence is exactly convergence in the norm:

$$\|f_n - f\|_\infty \longrightarrow 0.$$

Unlike pointwise convergence, uniform convergence controls every input with the same index.

Theorem 1.1. *The space $C(K)$ with the supremum norm is complete.*

Proof. Let (f_n) be Cauchy in the supremum norm. For each $x \in K$, the sequence $(f_n(x))$ is Cauchy in $\mathbb{C}$, so it has a limit $f(x)$. Given $\varepsilon > 0$, choose N such that $\|f_n - f_m\|_\infty < \varepsilon/2$ for $n, m \geq N$. Letting $m \to \infty$ at a fixed x gives $|f_n(x) - f(x)| \leq \varepsilon/2$. The bound holds for every x, so convergence is uniform.

The limit is continuous. Fix $a \in K$ and choose n with $\|f_n - f\|_\infty < \varepsilon/3$. Continuity of f_n at a gives $|f_n(x) - f_n(a)| < \varepsilon/3$ for $x \in K$ sufficiently close to a. Hence

$$|f(x) - f(a)| \leq |f(x) - f_n(x)| + |f_n(x) - f_n(a)| + |f_n(a) - f(a)| < \varepsilon.$$

Thus $f \in C(K)$ and $f_n \to f$ in the norm. ∎

For example, the geometric polynomials $p_N(x) = \sum_{n=0}^{N} x^n$ on $[0, r]$, with $0 < r < 1$, satisfy

$$\sup_{0 \leq x \leq r} \left| \frac{1}{1-x} - p_N(x) \right| \leq \frac{r^{N+1}}{1-r} \longrightarrow 0.$$

The same index controls the whole interval. On $[0, 1)$, the error is unbounded for each N, although convergence holds at every point. The distinction concerns the whole set on which convergence is asserted.

1.3 Uniform convergence and calculus

By Theorem 1.1, the uniform limit of continuous functions on K is continuous, since its unique pointwise limit is also its norm limit.

For $h = u + \mathrm{i}v$, define the complex Riemann integral by $\int h = \int u + \mathrm{i} \int v$ whenever both real integrals exist. The triangle inequality for finite Riemann sums passes to their limit.

Theorem 1.2 (Integration of a uniform limit). *Let $f_n : [a, b] \to \mathbb{C}$ be Riemann integrable, where $a < b$. If $f_n \to f$ uniformly, then f is Riemann integrable and*

$$\lim_{n \to \infty} \int_a^b f_n(x)\,\mathrm{d}x = \int_a^b f(x)\,\mathrm{d}x.$$

Proof. For real-valued functions, fix $\varepsilon > 0$ and choose n with $\sup_{x \in [a,b]} |f(x) - f_n(x)| < \varepsilon$. The function f is bounded because it is uniformly close to the bounded function f_n. If P is a partition of $[a, b]$, the upper Darboux sum $U(h, P)$ adds the subinterval lengths times the supremum of h on each subinterval. The lower sum $L(h, P)$ uses the infimum instead. For the functions f and f_n these sums satisfy

$$U(f, P) - L(f, P) \leq U(f_n, P) - L(f_n, P) + 2(b - a)\varepsilon.$$

Integrability of f_n makes its sum difference arbitrarily small by a suitable partition. Since ε is arbitrary, the Riemann criterion proves integrability of f. For complex-valued functions apply this argument to the real and imaginary parts. Finally,

$$\left| \int_a^b (f_n - f)(x)\,\mathrm{d}x \right| \leq (b - a) \sup_{x \in [a,b]} |f_n(x) - f(x)| \longrightarrow 0.$$ ∎

For a series, a summable bound supplies uniform convergence.

Theorem 1.3 (Weierstrass M-test). *Let S be a nonempty set and let $u_n : S \to \mathbb{C}$ satisfy $|u_n(x)| \leq M_n$ for every $x \in S$, where $M_n \geq 0$ and $\sum_n M_n < \infty$. Then $\sum_n u_n$ converges absolutely at each point and uniformly on S.*

Proof. The scalar comparison test gives absolute convergence at every point. The tail obeys $\sup_{x\in S} |\sum_{n>N} u_n(x)| \le \sum_{n>N} M_n \to 0$. This is uniform convergence. ∎

For the Gaussian series on $[0,1]$, each summand has absolute value at most $1/n!$. The real exponential series gives $e^{-x^2} = \sum_{n\ge 0}(-1)^n x^{2n}/n!$. The M-test and Theorem 1.2 allow termwise integration. Since $\int_0^1 x^{2n}\,\mathrm{d}x = 1/(2n+1)$, they prove Equation (1.1).

The differentiation theorem below assumes uniform convergence of the derivatives. Here $C^1[a,b]$ means differentiability on (a,b) with the function and its derivative extending continuously to the endpoints.

Theorem 1.4 (Differentiation of a limit)**.** *Let $f_n \in C^1[a,b]$. Suppose $f_n' \to g$ uniformly on $[a,b]$ and $f_n(x_0) \to c$ at one point $x_0 \in [a,b]$. Then $f_n \to f$ uniformly, where $f \in C^1[a,b]$ and $f' = g$ on (a,b).*

Proof. The uniform limit g is continuous. Define $f(x) = c + \int_{x_0}^{x} g(t)\,\mathrm{d}t$. The fundamental theorem of real calculus gives $f' = g$. Applying the same theorem to f_n gives

$$\|f_n - f\|_\infty \le |f_n(x_0) - c| + (b-a)\left\|f_n' - g\right\|_\infty \longrightarrow 0.$$ ∎

The value at one point is necessary. The constant functions $f_n(x) = n$ have derivatives converging uniformly to zero, but their values have no finite limit. Differentiation loses an additive constant.

1.4 Power series and smooth functions

For complex coefficients a_n, a centre $a \in \mathbb{C}$ and variable $z \in \mathbb{C}$, define the *radius of convergence* of $\sum_{n\ge 0} a_n(z-a)^n$ by

$$R = \sup\left\{ r \ge 0 : \sum_{n=0}^{\infty} |a_n|\, r^n < \infty \right\}.$$

The value $R = \infty$ is allowed. At the centre the series equals a_0. If $0 < r < R$, choose r_1 with $r < r_1 < R$ where the absolute series converges. Its terms are bounded, say $|a_n|\, r_1^n \le M$. For $|z-a| \le r$, the series and its derivative series are bounded by

$$|a_n(z-a)^n| \le M(r/r_1)^n, \qquad \left|n a_n (z-a)^{n-1}\right| \le \frac{M n r^{n-1}}{r_1^n}.$$

Both majorants are summable. If $|z-a| > R$, convergence would bound $|a_n|\,|z-a|^n$. The same geometric comparison would then give absolute convergence at a radius strictly between R and $|z-a|$, a contradiction. No claim is made about the boundary circle.

The complex derivative means the limit of $(f(z+h) - f(z))/h$ through complex increments $h \to 0$.

Theorem 1.5 (Differentiation of a power series)**.** *If $R > 0$, the sum $f(z) = \sum_{n\ge 0} a_n(z-a)^n$ has a complex derivative throughout $|z-a| < R$, given by*

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z-a)^{n-1}.$$

The series for f and every derivative converge uniformly on each concentric closed disk $|z-a| \le r < R$. In particular, $f^{(m)}(a) = m!\, a_m$ for every integer $m \ge 0$.

Proof. Let p_N be the partial sum and g the uniform limit of its derivatives on a smaller disk, as proved above. For a short segment inside that disk, polynomial differentiation and real calculus give

$$p_N(z+h) - p_N(z) = h \int_0^1 p_N'(z+th)\,\mathrm{d}t.$$

Pass to the limit using uniform convergence and Theorem 1.2. The continuous function g then satisfies $(f(z+h)-f(z))/h = \int_0^1 g(z+th)\,\mathrm{d}t \to g(z)$. For higher derivatives the majorants acquire only further polynomial factors in n. A polynomial times a geometric sequence with ratio less than one is summable, by the ratio test. The derived series therefore has radius at least R. Induction applies the same argument to every derived series, proving all the claims, including the coefficient formula at a. ∎

For example, the geometric series gives $f(z) = 1/(1-z)$ for $|z| < 1$. Its difference quotient is $(1-z-h)^{-1}(1-z)^{-1}$, which tends to $(1-z)^{-2}$. The theorem therefore gives $f'(z) = \sum_{n\ge 1} n z^{n-1} = 1/(1-z)^2$ there. Neither equality asserts convergence at $z = 1$.

A real function is *real-analytic* near a point if it equals a convergent real power series centred at that point, on some neighbourhood. Smoothness, meaning continuous derivatives of every order, is weaker. Consider

$$g(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Away from zero, its derivatives have the form $g^{(m)}(x) = P_m(1/x)e^{-1/x^2}$, where P_m is a polynomial. The chain rule proves this inductively from

$$P_0(y) = 1, \qquad P_{m+1}(y) = -y^2 P_m'(y) + 2y^3 P_m(y).$$

Every power of $1/|x|$ times e^{-1/x^2} tends to zero as $x \to 0$. Indeed, $e^{y^2} \ge y^{2j}/j!$ for $y > 0$, and j can be chosen larger than half any specified power. Difference quotients at zero now prove inductively that all derivatives exist there and equal zero. The same bounds give their continuity. If g equalled a power series $\sum_m a_m x^m$ near zero, the radius argument above would give positive radius. The real-increment case of Theorem 1.5 would give $a_m = g^{(m)}(0)/m! = 0$ for every m. Yet $g(x) > 0$ for $x \neq 0$. Thus g is smooth and is not real-analytic at zero.

1.5 The contraction principle

A *fixed point* of a map $T : X \to X$ is a point u_* with $Tu_* = u_*$. A map is a *contraction* if distances shrink by a factor bounded by one constant $0 \le q < 1$. A subset of a normed space is *closed* if it contains the limit in the ambient space of every convergent sequence from the subset.

Theorem 1.6 (Banach contraction principle). *Let X be a nonempty closed subset of a Banach space and let $T : X \to X$ satisfy*

$$\|Tu - Tv\| \le q\,\|u - v\|, \qquad u, v \in X, \quad 0 \le q < 1.$$

Then T has exactly one fixed point u_*. For every $u_0 \in X$, the iteration $u_{n+1} = Tu_n$ converges to u_*. With $q^0 = 1$,

$$\|u_n - u_*\| \le \frac{q^n}{1-q}\,\|u_1 - u_0\|, \qquad n \ge 0. \tag{1.2}$$

Proof. Successive increments satisfy $\|u_{n+1} - u_n\| \le q^n \|u_1 - u_0\|$ by induction. For $p \ge 1$, the triangle inequality and a finite geometric sum give

$$\|u_{n+p} - u_n\| \le \frac{q^n(1-q^p)}{1-q} \|u_1 - u_0\|.$$

Hence (u_n) is Cauchy. Completeness of the ambient space gives a limit, and closedness puts it in X. The contraction bound makes T continuous, so the limit satisfies $Tu_* = u_*$. If v_* is another fixed point, then $\|u_* - v_*\| \le q \|u_* - v_*\|$, forcing equality of the points. Letting $p \to \infty$ in the bound proves Equation (1.2). ∎

For example, the contraction $T(u) = 1 + u/2$ on $\mathbb{R}$ has fixed point 2. Starting with $u_0 = 0$ gives

$$u_n = 2(1 - 2^{-n}), \qquad |u_n - 2| = 2^{1-n}.$$

The error equals the bound in Equation (1.2) with $q = 1/2$.

Kepler's equation for an elliptic orbit is $E - e \sin E = M$, with $M \in \mathbb{R}$ and $0 \le e < 1$. Here e is the orbital eccentricity, E is the eccentric anomaly and M is the mean anomaly. The letter e elsewhere denotes Euler's number. The map $T(E) = M + e \sin E$ maps $\mathbb{R}$ to itself and has derivative bounded in absolute value by e. The real mean value theorem therefore makes it a contraction. There is a unique solution for every M.

1.6 The pendulum as an integral equation

The pendulum's angle x and angular velocity v obey

$$\dot{x} = v, \qquad \dot{v} = -\nu_0^2 \sin x,$$

where $\nu_0 > 0$ is its small-oscillation frequency. We use ν for frequency because ω is reserved for the collection's symplectic form. Writing $u = (x, v)$ and $F(x, v) = (v, -\nu_0^2 \sin x)$, the initial-value problem has the integral form

$$(\mathcal{P}u)(t) = u_0 + \int_0^t F(u(s))\, \mathrm{d}s, \qquad u = \mathcal{P}u.$$

Fix a time bound $T > 0$. On the path space $X = C([-T, T], \mathbb{R}^2)$, use the scaled norm

$$\|u\|_X = \sup_{|t| \le T} \max\{|x(t)|, |v(t)| / \nu_0\}.$$

This path space is complete by the proof of Theorem 1.1, coordinate by coordinate. Since $|\sin x - \sin y| \le |x - y|$,

$$\|\mathcal{P}u - \mathcal{P}w\|_X \le T\nu_0 \|u - w\|_X.$$

Thus the contraction principle applies for $T\nu_0 < 1$. The real fundamental theorem of calculus turns its fixed point into a solution of the differential equations. Lecture 2 develops the general theory of existence and uniqueness from this construction.

Summary.

- The space $C(K)$ is complete in the supremum norm for nonempty compact K.
- Uniform limits of continuous functions are continuous. Uniform limits of Riemann integrable functions on a bounded interval preserve their integrals. Uniform convergence of the derivatives, with convergence at one point, justifies differentiating the limit.

- A power series and its derivatives converge uniformly on smaller concentric closed disks inside its radius of convergence. Its sum is complex-differentiable there, with the termwise derivative.
- A contraction on a nonempty closed subset of a Banach space has a unique fixed point, with the iteration error bounded by Equation (1.2).

Lecture 2 applies completeness and contraction to differential equations.

References

R. S. Strichartz, *The Way of Analysis*, revised edition, 2000, §§7.3–7.4 and §9.3.4. W. Rudin, *Principles of Mathematical Analysis*, third edition, 1976, Chapter 7. V. A. Zorich, *Mathematical Analysis II*, second edition, 2016, Chapter 16 and §9.7.